Synthetic Data Worksheet

GLSS6 2012/13 → DHS 2014 synthetic data generation | IndabaX Ghana 2026 (ACET)

Name / Group:

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The printed codebook defines every variable below.

Part A — Sort the variables (Think–Pair–Share)

Every variable plays one of **three** roles.

First, **write one question** you could *only* answer by combining both surveys—name the DHS outcome and the GLSS-only variable it needs.

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Second, write the role for each variable in the blank column.

Roles: (X) Shared predictor X — observed in *both* surveys (Y) Synthetic target — observed in *GLSS only*, generated for DHS (O) Analysis outcome — a *DHS child* outcome studied after synthetic data generation

#	Variable	Role (X/Y/O)	#	Variable	Role (X/Y/O)
1	region		9	total household expenditure	
2	welfare (per adult eq.)		10	NHIS registration (any member)	
3	child stunting		11	cooking fuel	
4	has electricity		12	DPT3 vaccination	
5	household size		13	health spending (`TOTHLTH`)	
6	poverty status		14	water source	
7	child fever (last 2 wks)		15	toilet type	
8	head age		16	child diarrhea (last 2 wks)	

Part B — Factor the model (Pairs)

Three variable types are in play (from Part A):

- **shared predictors** X — region, urban, household size, ... (observed for every DHS household)
- **synthetic targets** — **log welfare** and **NHIS registration**: the two we generate in the practical
- a **derived** variable — poverty status

1. Write down the joint model. Write the joint model of the two synthetic targets as a chain of conditional draws. *Hint: condition the second target on the first.*

$$p(\log(\text{welfare}),\ \text{NHIS} \mid X) \ =$$

2. Special case: welfare only. Suppose we synthesize only welfare and derive poverty from the welfare draw. What does the model become?

$$p(\log(\text{welfare}) \mid X) \ =$$

3. Discuss. Why might we synthesize a block of multiple targets instead of only welfare? Write one reason connected to the practical.